

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mining Engineering) SEMESTER - 3 Summer 2026 ( Supply. )

Course : Bachelor of Technology (Mining Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1000BS301 - ENGINEERING MATHEMATICS - III

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. The Laplace transform of  $f(s) = \frac{3s+4}{s^2+9}$  is equal to
  - a)  $3 \cos 3t - \frac{4}{3} \sin 3t$
  - ☒ b)  $3 \cos 3t + \frac{4}{3} \sin 3t$
  - c)  $3 \cos 3t + \sin 3t$
  - d) None
2. The Laplace transform of  $F(t) = e^{-3t} \sin 2t$  is equal to
  - a)  $\frac{2}{(s+3)^2-4}$
  - ☒ b)  $\frac{2}{(s+3)^2+4}$
  - c)  $\frac{2}{(s-3)^2-4}$
  - d)  $\frac{2}{(s+3)^2+4}$
3. The inverse Laplace transform of  $\log\left(\frac{s+1}{s-1}\right)$  is given by
  - a)  $2 \frac{\cosh t}{t}$
  - b)  $2t \cos t$
  - ☒ c)  $2 \frac{\sinh t}{t}$
  - d)  $2t \sin t$
4. If  $F\{f(x)\} = F(s)$ , then by parseval's identity  $\frac{1}{2\pi} \int_{-\infty}^{\infty} [F(s)]^2 ds$  is equal to
  - ☒ a)  $\int_{-\infty}^{\infty} [F(x)]^2 dx$
  - b)  $\frac{1}{\pi} \int_0^{\infty} [F(x)]^2 dx$
  - c)  $2\pi \int_0^{\infty} [F(x)]^2 dx$
  - d) None
5. The inverse Fourier cosine transform is
  - a)  $f(x) = \int_0^{\infty} F_c(s) \sin(sx) ds$
  - b)  $f(x) = \frac{2}{\pi} \int_{-\infty}^{\infty} F_c(s) \cos(sx) ds$
  - ☒ c)  $f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(s) \cos(sx) ds$
  - d)  $f(x) = \frac{2}{\pi} \int_{-\infty}^{\infty} F_c(s) \sin(sx) ds$
6. If  $\{F_c(ax)\} = kF_c\left(\frac{s}{a}\right)$ , then  $k$  is equal to
  - a)  $\frac{2}{a}$
  - b)  $a$
  - c)  $\frac{a}{2}$
  - ☒ d)  $\frac{1}{a}$

- 7 The solution of  $p + q = z$
- a)  $f(x + y, y + \log z) = 0$       b)  $f(x^2 + y^2 + z^2, xyz) = 0$
- c)  $f(xy, y \log z) = 0$       ~~d)  $f(x - y, y - \log z) = 0$~~
- 8 A solution of  $u_{xy} = 0$  is
- a)  $u = \int f(y) dy + \varphi(x)$       b)  $u = \int f(x) dx + \varphi(y)$
- c)  $u = \int f(y) dy + \varphi(y)$       d) None
- 9 The value of  $\int_0^\infty f(t) \delta(t - a) dt$  is
- a) 1      b) 0
- c)  $f(a)$       d)  $f(0)$
- 10 The value of  $\int \frac{dz}{z-1}$  over the  $C$  being  $|z| = 2$ , is
- a) 0      b)  $2\pi i$
- c)  $\pi i$       d) None
- 11 The singular points of  $\frac{\cos \pi z}{(z-1)(z-2)}$
- a) 0, 1      b) 1, 2
- c) -1, -2      d) 0, 2
- 12 The Laplace equation is
- a)  $u_{xx} + u_{yy} = 0$       b)  $u_{xx} + u_{yy} + u_{zz} = 0$
- c)  $u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0$       d) All of the above
- Q2. Solve the following.
- A) Find the Laplace Transform of  $e^{4t} \cos^3 t$ . 6
- B) Evaluate  $\int_0^\infty \frac{\cos at - \cos bt}{t} dt$  by using Laplace transform. 6
- Q3. Solve the following.
- A) Find the inverse Laplace Transform of  $\log\left(\frac{s+a}{s+b}\right)$ . 6
- B) Solve using Laplace transform  $3 \frac{dy}{dt} + 2y = e^{3t}$ ,  $y = 1$  at  $t = 0$ . 6
- Q4. Solve Any Two of the following.
- A) Find the Fourier Transform of  $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$  and hence evaluate  $\int_0^\infty \frac{\sin x}{x} dx$ . 6
- B) Find Fourier cosine transform of  $f(x) = e^{-x} + e^{-2x}$ ,  $x > 0$ . 6
- C) Using Parseval's identity, prove that  $\int_0^\infty \frac{t^2}{(t^2+a^2)(t^2+b^2)} dt = \frac{\pi}{2(a+b)}$ . 6
- Q5. Solve Any Two of the following.
- A) Form the partial differential equation by eliminating arbitrary function  $f$  from  $f(x + y + z, x^2 + y^2 + z^2) = 0$ . 6

B) Solve  $pz - qz = z^2 + (x + y)^2$ . 6

C) Use the method of separation of variables to solve the equation  $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$ , given that  $u(x, 0) = 6e^{-3x}$ . 6

Q6. Solve Any Two of the following.

A) Determine the analytic function  $f(z)$  in terms of  $z$  whose real part is  $x^2 - y^2$ . 6

B) Prove that  $u = x^2 - y^2 - 2xy - 2x + 3y$  is harmonic. Find a function  $v$  such that  $f(z) = u + iv$  is analytic. 6

C) Evaluate  $\oint \frac{e^{2z}}{(z+1)^4} dz$  where  $c$  is the circle  $|z| = 2$ . 6

\* \* \* End \* \* \*